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# Sang Hyeok Im (임상혁)

## Curriculum Vitae

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea — sanghyeok.im97@gmail.com

## EDUCATION

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### Ph.D student in Astronomy Program (2021.02 - present)

Department of Physics and Astronomy, Seoul National University, 1, Gwanak-ro, Gwanak-gu, Seoul, Republic of Korea

### B.S. in Physics (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

### B.S. in Astronomy and Atmospheric Sciences (2015.03 - 2021.02; double major)

Kyungpook National University, 80, Daehak-ro, Buk-gu, Daegu, 41566, Republic of Korea

## RESEARCH INTERESTS

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1. Large-scale structures of the universe
2. Observational cosmology
3. Environmental effects on galaxy properties
4. Formation and evolution of proto-clusters

## PUBLICATIONS

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### Refereed Journal Article

#### First Author

1. Testing Ly $\alpha$  Emitters and Lyman-Break Galaxies as Tracers of Large-Scale Structures at High Redshifts  
**Sang Hyeok Im** et al.  
2024, ApJ, 972, 196 (NASA ADS)  
**Highlighted as AAS Nova**<sup>1</sup> (see here)
2. ODIN: Rest-frame Optical Morphologies and Star Formation Activity of Ly $\alpha$  Emitters at  $z = 2.4, 3.1$ , and 4.5  
**Sang Hyeok Im** et al.,  
2026, submitted to ApJ (arXiv:2605.29344)

#### Second Author

1. Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies  
Gene Yun, **Sang Hyeok Im** et al.,  
2026, accepted for publication in ApJ (arXiv:2607.17271)

#### Co-Author

1. ODIN: Searching for LyC emission from Lyman- $\alpha$  emitters at  $z = 4.5$  in the E-COSMOS and XMM-LSS fields  
Eunsuk Seo et al., (**Sang Hyeok Im**)  
2026, ApJ, 1001, 224 (NASA ADS)
2. ODIN: Clustering Properties of Ly $\alpha$  Blobs at  $z \sim 2.4$  and 3.1  
Byeongha Moon et al., (**Sang Hyeok Im**)

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<sup>1</sup>AAS Nova is the research highlights from the journals of the American Astronomical Society.

2026, ApJ, 1004, 122 (NASA ADS)

3. ODIN: A New Lyman Alpha Blob Selection Method, Sample, and Statistical Analysis at  $z \sim 3.1$   
Byeongha Moon et al., (**Sang Hyeok Im**)  
2025, ApJ, 998, 345 (NASA ADS)
4. ODIN: Characterizing the Three-dimensional Structure of Two Protocluster Complexes at  $z=3.1$   
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)  
2025, submitted to ApJ (arXiv:2511.11826)
5. ODIN: Star Formation Histories Reveal Formative Starbursts Experienced by Ly $\alpha$ -emitting Galaxies at Cosmic Noon  
Nicole M. Firestone et al., (**Sang Hyeok Im**)  
2025, ApJL, 986, 8 (NASA ADS)
6. ODIN: High Clustering Strength of Protoclusters at Cosmic Noon  
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)  
2025, ApJ, 982, 74 (NASA ADS)
7. ODIN: Identifying Protoclusters and Cosmic Filaments Traced by Ly $\alpha$ -emitting Galaxies  
Vandana Ramakrishnan et al., (**Sang Hyeok Im**)  
2024, ApJ, 977 119 (NASA ADS)
8. ODIN: Improved Narrowband Ly $\alpha$ Emitter Selection Techniques for  $z = 2.4, 3.1$ , and 4.5  
Nicole M. Firestone et al., (**Sang Hyeok Im**)  
2024, ApJ, 974, 217 (NASA ADS)
9. The One-hundred-deg<sup>2</sup> DECam Imaging in Narrowbands (ODIN): Survey Design and Science Goals  
Kyoung-Soo Lee et al., (**Sang Hyeok Im**)  
2024, ApJ, 962, 36 (NASA ADS)
10. ODIN: Where Do Ly $\alpha$  Blobs Live? Contextualizing Blob Environments within the Large-Scale Structure  
Vandana Ramakrishnan, Byeongha Moon, **Sang Hyeok Im** et al.,  
2023, ApJ, 951, 119 (NASA ADS)

## GRANTS AND FELLOWSHIPS

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### CIAS Student Fellow (2021.09 - 2026.02)

- o Institutional Location: Korea Institute for Advanced Study, 85 Hoegi-ro, Dongdaemun-gu, Seoul 02455, Republic of Korea
- o Project Title: A 100 Deg<sup>2</sup> DECam Narrow-band Survey for the LSST Era: Tracing the Largest Cosmic Structures in the Distant Universe

## PRESENTATIONS

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### Oral Presentations

1. **110<sup>th</sup> KAS Fall Meeting**  
Date: 2024 Oct. 16 - 18  
Location: Gangneung-si, Korea  
Title: Testing LAEs and LBGs as Tracers of Large-Scale Structures at High Redshifts
2. **K-GMT Science Program Users Meeting 2024**  
Date: 2024 Aug. 21 - 23  
Location: KASI, Daejeon, Korea  
Title: Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies
3. **9<sup>th</sup> Galaxy Evolution Workshop**  
Date: 2023 Feb. 20 - 23

Location: Kyoto University, Japan

Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in the high-redshift universe

## Poster Presentations

1. **ELT Science in Light of JWST**

Date: 2024 June 3 - 7

Location: Tohoku University, Sendai, Japan

Title: Large-scale structures in the universe traced by Lyman alpha emitters and Lyman-break galaxies

2. **108<sup>th</sup> KAS Fall Meeting**

Date: 2023 Oct. 18 - 20

Location: Ramada Plaza Jeju, Jeju, Republic of Korea

Title: Testing Lyman alpha emitters and Lyman-break galaxies as tracers of the large-scale structures in the high-z universe

3. **XXXI<sup>th</sup> General Assembly International Astronomical Union**

Date: 2022 Aug. 2 - 11

Location: BEXCO, Busan, Republic of Korea

Title: Lyman alpha emitters and Lyman-break galaxies as tracers of large-scale structures in a simulated universe

## RESEARCH EXPERIENCES

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### Observing Proposal (PI)

1. **Gemini/N GMOS-longslit (2023B)**

**Title:** Studying the Ongoing Morphological Transformation from Elliptical to Spiral Galaxies Induced by Gas Accretion from Neighboring Galaxies

**Allocated Time:** 10.1 hr

Results are submitted to ApJ (as a second author)

### Observing Proposal (CoI)

1. **MMT/Hectospec (2024A - 2026A)**

**Title:** Revisiting the Faint-end Cutoff in the Galaxy Luminosity Function of the Satellites around Isolated Early-type Galaxies

**Allocated Time:** total 7 nights

References Section

## REFERENCES

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**Prof. Ho Seong Hwang**

Seoul National University, Korea

galaxy79@snu.ac.kr